

- 8 (a) *Petunia axillaris* is a plant with flowers that are pollinated by night-flying moths. To attract moths, *P. axillaris* synthesises compounds at night that are released as scents (attractive smells).

Figure 8.1 shows a flower of *P. axillaris*.

Figure 8.2 shows a moth with large sensory structures known as antennae.



Figure 8.1



Figure 8.2

- (i) Moths detect scent chemicals using sensory receptor cells on their antennae.

Outline the role of sensory receptor cells.

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- (ii) The compounds that are released as scents by *P. axillaris* are lipids.

Outline how *P. axillaris* could control the production of scent chemicals by regulating gene expression, so that scent chemicals are synthesised **only** at night.

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(b) Microarrays could be used to investigate whether gene expression in *P. axillaris* is different during the day compared to the night.

Figure 8.3 shows some steps that would be performed in this microarray analysis.

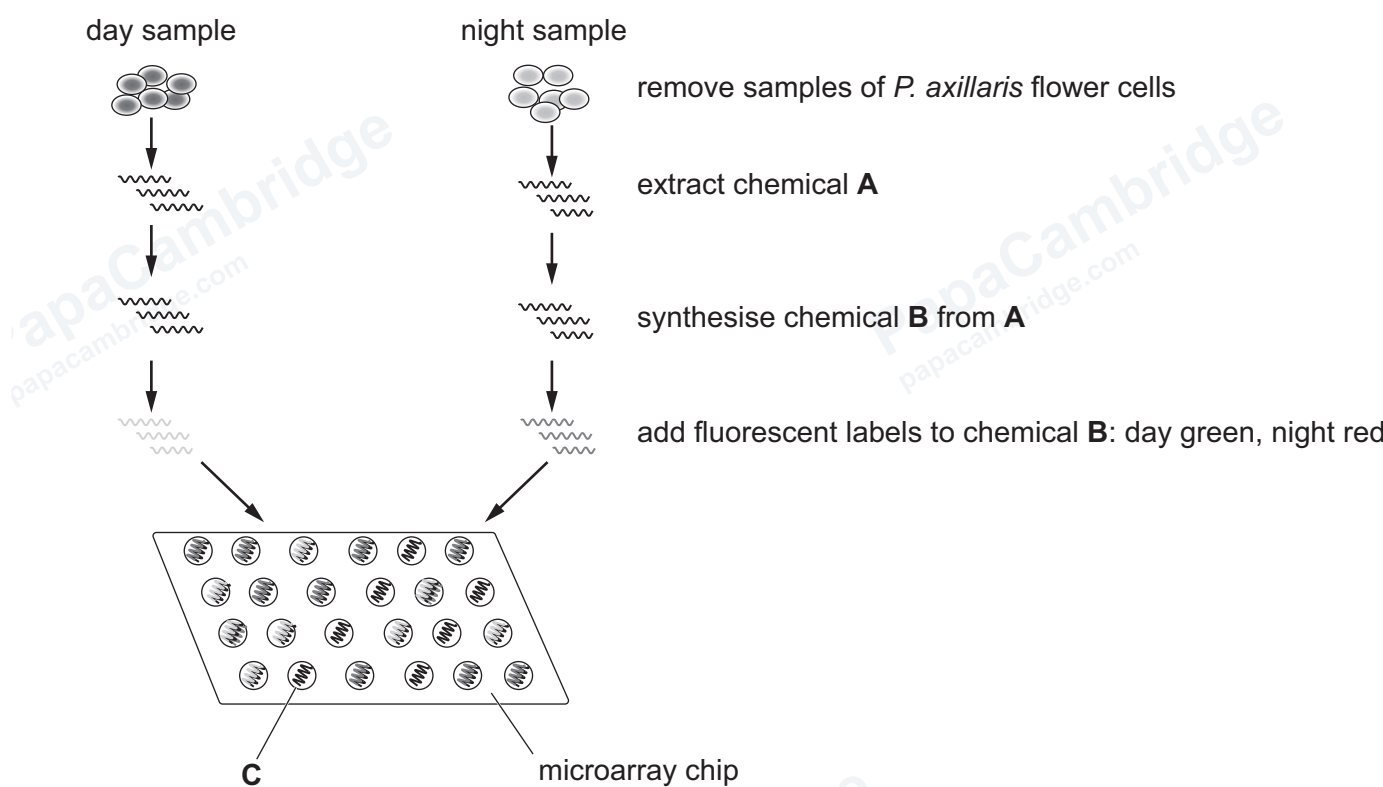


Figure 8.3

Identify **A**, **B** and **C** on Figure 8.3.

A

B

C [3]

[Total: 9]